

SPECTRAL KAKUTANI–FELDMAN–HÁJEK CRITICALITY OF CORRELATED CONFORMATIONAL MODES: MODE AMPLITUDE PERTURBATION, ISO-SPECTRAL GIVENS ROTATION, AND TAIL-INCREMENT FINITE-SIZE ELIMINATION

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SUMMARY. Paper II of this series [6] decomposed the covariance part of the correlated Kakutani index of two Gaussian conformational ensembles into a commuting amplitude part and a non-commuting rotation part, $D_{\text{cov}} = D_{\text{comm}} + D_{\text{rot}}$, and closed with the question of what happens when a ligand's effect on the collective modes decays along the mode index. This paper answers it. Two spectral perturbation mechanisms are defined on the cone $\mathcal{P}_N$ of positive definite matrices: *mode amplitude perturbation* (Mechanism A), $\lambda_i^{(B)} = \lambda_i^{(A)}(1 + c i^{-\alpha_\lambda})$ in a common eigenbasis, whose exact per-mode contribution is $\frac{1}{2} \ln \frac{1+\delta/2}{\sqrt{1+\delta}} = \sum_{k \geq 2} a_k \delta^k$ with $a_k = (-1)^k (1-2^{1-k})/(4k)$; and *iso-spectral Givens rotation* (Mechanism B), $\Lambda_A = \Lambda_B$ with the k -th adjacent pair of modes of anisotropy ratio r rotated by $\theta_k = c_\theta k^{-\alpha_\theta}$, whose exact per-pair contribution is $\frac{1}{2} \ln[1 + \kappa(r) \sin^2 \theta_k]$, $\kappa(r) = (r-1)^2/(4r)$. Both are normalised to the leading coefficient $A = 0.0100$ of Paper I. Theorem 3.1 gives the unified three-regime Euler–Maclaurin asymptotics of $D_{\text{cov}}(N)$ for both mechanisms, with the limit constants in closed form, and the Feldman–Hájek dichotomy at $\alpha_c = \frac{1}{2}$ for each mechanism separately; the predictions agree with the numerical experiment to 3×10^{-12} or better at $N \geq 100$ and to 10^{-8} at $N = 10$. Theorem 4.1 proves that the doubling tail increment $D_{\text{tail}}(N) = D_{\text{cov}}(2N) - D_{\text{cov}}(N)$ annihilates the Euler–Maclaurin constant $A\zeta(2\alpha)$, so that its regression exponent $\gamma_{\text{tail}}(\alpha)$ equals $1 - 2\alpha$ on the whole range $0 < \alpha < 1$, including the negative subcritical values, and $\gamma_{\text{tail}}^+ = \max(\gamma_{\text{tail}}, 0)$ is the sharp phase curve; the bias of the naive exponent at $\alpha = 0.45$ falls from $+0.073$ to $+0.003$ (A) and from $+0.062$ to $+0.0001$ (B). Theorem 4.2 gives the refined finite-size formula that reproduces the naive exponent to four decimals (0.1729 at $\alpha = 0.45$, where Paper I's two-term formula gives 0.1601). Theorem 4.3 explains the residual difference between the two mechanisms by the geometry of $\mathcal{P}_N$: in the geodesic coordinate $\eta_i = \ln(\lambda_i^{(B)}/\lambda_i^{(A)})$ the per-mode divergence $\frac{1}{2} \ln \cosh(\eta_i/2)$ is even, the odd terms of Mechanism A are the curvature of the Euclidean coordinate $\delta_i = e^{\eta_i} - 1$, and the resulting shift $\Delta\gamma_{\text{tail}} = \alpha c \frac{1-2\alpha}{1-3\alpha} \frac{2^{1-3\alpha}-1}{2^{1-2\alpha}-1} N^{-\alpha}$ is verified; the geodesic variant $\lambda_i^{(B)} = \lambda_i^{(A)} e^{c i^{-\alpha_\lambda}}$ and the Givens mechanism, both even, give $\gamma_{\text{tail}}(0.45) = +0.1001$. Theorem 5.1 establishes the joint $(\alpha_\lambda, \alpha_\theta)$ phase diagram: the infinite ensembles are equivalent iff $\min(\alpha_\lambda, \alpha_\theta) > \frac{1}{2}$, with effective exponent $\max(1 - 2\min(\alpha_\lambda, \alpha_\theta), 0)$, confirmed on a 19×19 grid with no misclassified point, and Corollary 5.3 gives the finite- N exponent on that grid as the Boltzmann-type convex combination of two power laws, reproducing all 361 measured values to 4×10^{-4} ; the rotation-only singular phase is the iso-spectral signature of cryptic allostery. The paper closes with an assessment and with the empirical-covariance (Marchenko–Pastur) problem for the next step.

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1. INTRODUCTION AND SUMMARY OF MAIN RESULTS

What Papers I and II established. Paper I [5] modelled a protein with N independent two-state switches and two ligands as product Bernoulli measures, defined the marginal drug Kakutani index $K_N = \sum_{i \leq N} h_i$ (a sum of local squared Hellinger distances), and proved for the power-law shift $q_i = p + ci^{-\alpha}$ the trichotomy: $K_N \sim AN^{1-2\alpha}/(1-2\alpha)$ for $\alpha < \frac{1}{2}$, $K_N = A(\ln N + \gamma_{\text{EM}}) + O(1)$ at $\alpha = \frac{1}{2}$, $K_N \rightarrow K_\infty < \infty$ for $\alpha > \frac{1}{2}$, with $A = c^2/(4p(1-p)) = 0.0100$; by Kakutani’s theorem [9] the infinite ensembles are singular in the first two cases and equivalent in the third. It also isolated a finite-size artefact: the exponent read from a regression of $\ln K_N$ on $\ln N$ over $[10^3, 10^5]$ is $\gamma_{\text{eff}} = (1-2\alpha)/(1+(1-2\alpha)\zeta(2\alpha)N^{2\alpha-1})$, and because $\zeta(2\alpha) \rightarrow -\infty$ as $2\alpha \rightarrow 1$ the measured exponent at $\alpha = 0.45$ is 0.161 instead of 0.100; the doubling estimator $K_{2N} - K_N$ removed the bias. Paper II [6] replaced the product measures by Gaussian ensembles $\mathcal{N}(\mu_A, C_A)$, $\mathcal{N}(\mu_B, C_B)$, defined the correlated index as the Bhattacharyya distance $D_B = D_{\text{mean}} + D_{\text{cov}}$, proved the commuting–rotation decomposition

$$D_{\text{cov}} = D_{\text{comm}} + D_{\text{rot}}, \quad D_{\text{comm}} = \frac{1}{2} \sum_{i=1}^N \ln \cosh \frac{\eta_i}{2}, \quad \eta_i = \ln \frac{\lambda_i^{(B)}}{\lambda_i^{(A)}}, \quad D_{\text{rot}} \geq 0, \quad (1.1)$$

with $D_{\text{rot}} = 0$ iff the covariances share an ordered eigenbasis, and showed for Kac–Murdock–Szegő networks that the covariance term dominates the mean term (99.95%) and diverges linearly in N because the network change touches every mode with undiminished strength.

The three questions of this paper. In a real protein an allosteric ligand reshapes strongly only the low-frequency collective modes; its effect on the high-frequency, localised modes decays with the mode index. This raises three questions.

- (1) Do the two parts of (1.1)—a perturbation of the mode amplitudes (Mechanism A) and an iso-spectral rotation of the mode directions (Mechanism B, $\Lambda_A = \Lambda_B$)—each induce a Kakutani–Feldman–Hájek transition, and is the critical exponent again $\alpha_c = \frac{1}{2}$?
- (2) Can a difference operator remove the Euler–Maclaurin constant $\zeta(2\alpha)$ at its root, so that at finite $N \leq 10^5$ the phase curve is the sharp kink $\max(1-2\alpha, 0)$ and, at the same time, the negative tail exponent $1-2\alpha < 0$ of the subcritical regime is measured?

(3) Why, at $\alpha = 0.45$, is the doubling-tail exponent of Mechanism B equal to $+0.1001$ (error 10^{-4}) while that of Mechanism A is $+0.1028$ (shift $+0.0028$)?

The answers are: yes, with $\alpha_c = \frac{1}{2}$ for each mechanism separately (Theorem 3.1); yes, the doubling tail increment (Theorem 4.1), whose remaining finite-size residual is fully predicted by Theorem 4.2; and because Mechanism A is written in the Euclidean coordinate $\delta_i = \lambda_i^{(B)}/\lambda_i^{(A)} - 1$ of the Riemannian symmetric space $\mathcal{P}_N = \text{GL}_N(\mathbb{R})/\text{O}(N)$, whereas the Bhattacharyya divergence is an even function of the geodesic coordinate $\eta_i = \ln(1 + \delta_i)$; the odd terms are the curvature of the pull-back, and they vanish for the geodesic variant of Mechanism A and for the Givens rotation, whose angle is already a geodesic coordinate in $\text{SO}(2)$ (Theorem 4.3). Theorem 5.1 combines the two mechanisms into a joint phase diagram whose rotation-only singular quadrant is the mathematical form of cryptic, iso-spectral allostery.

Standing notation. $\mathcal{P}_N$ is the cone of symmetric positive definite $N \times N$ matrices. For $C_A, C_B \in \mathcal{P}_N$, $S = \frac{1}{2}(C_A + C_B)$ and

$$D_{\text{cov}} = \frac{1}{2} \ln \det S - \frac{1}{4} \ln \det C_A - \frac{1}{4} \ln \det C_B.$$

$H_N(s) = \sum_{i \leq N} i^{-s}$; ζ is the Riemann zeta function (meromorphic, simple pole at 1); $\gamma_{\text{EM}} = 0.5772156649 \dots$. The regression window is $[10^3, 10^5]$ with a sixty-point log-uniform grid and logarithmic centre $N_* = 10^4$. The benchmark parameters are $c = 0.40$ for Mechanism A (so that $c^2/16 = 0.0100$), $r = 3$ and $c_\theta = \sqrt{0.06} = 0.2449490$ for Mechanism B (so that $\kappa(3)c_\theta^2/2 = 0.0100$), and $A = 0.0100$ denotes the common leading coefficient.

2. SPECTRAL AND ROTATIONAL PERTURBATION MECHANISMS ON THE SPD CONE

Definition 2.1 (Mechanism A: collective mode amplitude scaling). Let $C_A = Q\Lambda_A Q^\top$ with $\Lambda_A = \text{diag}(\lambda_1^{(A)} \geq \dots \geq \lambda_N^{(A)} > 0)$, and let $C_B = Q\Lambda_B Q^\top$ have the same eigenvectors with

$$\lambda_i^{(B)} = \lambda_i^{(A)}(1 + \delta_i), \quad \delta_i = c i^{-\alpha_\lambda}, \quad 0 < c < 1, \quad \alpha_\lambda > 0. \quad (2.1)$$

Then $D_{\text{rot}} = 0$ and $D_{\text{cov}}(N) = D_{\text{comm}}(N) = \sum_{i \leq N} d_{\text{cov},i}$ with $d_{\text{cov},i} = \frac{1}{2} \ln \frac{\lambda_i^{(A)} + \lambda_i^{(B)}}{2\sqrt{\lambda_i^{(A)}\lambda_i^{(B)}}}$. The *geodesic variant* (Mechanism A-geo) replaces (2.1) by $\lambda_i^{(B)} = \lambda_i^{(A)} e^{c i^{-\alpha_\lambda}}$, i.e. $\eta_i = c i^{-\alpha_\lambda}$.

Proposition 2.2 (Exact per-mode contribution of Mechanism A). For $\delta \in (-1, 1)$,

$$d_{\text{cov}}(\delta) = \frac{1}{2} \ln \frac{1 + \delta/2}{\sqrt{1 + \delta}} = \sum_{k \geq 2} a_k \delta^k, \quad a_k = \frac{(-1)^k (1 - 2^{1-k})}{4k}, \quad (2.2)$$

so that

$$d_{\text{cov}}(\delta) = \frac{\delta^2}{16} - \frac{\delta^3}{16} + \frac{7\delta^4}{128} - \frac{3\delta^5}{64} + \frac{31\delta^6}{768} - \frac{9\delta^7}{256} + O(\delta^8). \quad (2.3)$$

The series converges absolutely for $|\delta| < 1$, and $|a_k| \leq 1/(4k)$.

Proof. With $\lambda^{(B)} = \lambda^{(A)}(1 + \delta)$, $\frac{\lambda^{(A)} + \lambda^{(B)}}{2\sqrt{\lambda^{(A)}\lambda^{(B)}}} = \frac{2 + \delta}{2\sqrt{1 + \delta}} = \frac{1 + \delta/2}{\sqrt{1 + \delta}}$, which gives the closed form. Expanding, $\ln(1 + \delta/2) = \sum_{k \geq 1} (-1)^{k+1} \delta^k / (k 2^k)$ and $\frac{1}{2} \ln(1 + \delta) = \sum_{k \geq 1} (-1)^{k+1} \delta^k / (2k)$ for $|\delta| < 1$, so

$$d_{\text{cov}}(\delta) = \frac{1}{2} \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \left(\frac{1}{2^k} - \frac{1}{2} \right) \delta^k = \sum_{k \geq 1} \frac{(-1)^k (2^{-1} - 2^{-k})}{2k} \delta^k = \sum_{k \geq 1} \frac{(-1)^k (1 - 2^{1-k})}{4k} \delta^k.$$

The $k = 1$ coefficient is 0, and $a_2 = \frac{1-1/2}{8} = \frac{1}{16}$, $a_3 = -\frac{1-1/4}{12} = -\frac{1}{16}$, $a_4 = \frac{1-1/8}{16} = \frac{7}{128}$, $a_5 = -\frac{1-1/16}{20} = -\frac{3}{64}$, $a_6 = \frac{1-1/32}{24} = \frac{31}{768}$, $a_7 = -\frac{1-1/64}{28} = -\frac{9}{256}$. Absolute convergence for $|\delta| < 1$ and the bound follow from $0 < 1 - 2^{1-k} < 1$. $\square$

On the coefficients. The series (2.2) is alternating with closed-form coefficients $a_2 = \frac{1}{16}$, $a_3 = -\frac{1}{16}$, $a_4 = \frac{7}{128}$, $a_5 = -\frac{3}{64}$, $a_6 = \frac{31}{768}$, $a_7 = -\frac{9}{256}$; the companion script confirms them independently, the residual $(d_{\text{cov}} - \sum_{k \leq 4} a_k \delta^k)/\delta^5$ tending to $-0.046875 = -\frac{3}{64}$ as $\delta \rightarrow 0$. The ratio $|a_{k+1}/a_k| = \frac{k}{k+1} \frac{1-2^{-k}}{1-2^{1-k}} \rightarrow 1$, so the radius of convergence is exactly 1: it is set by the algebraic branch point of $\sqrt{1+\delta}$ at $\delta = -1$, not by $\ln(1+\delta/2)$, whose singularity lies at $\delta = -2$. With $c = 0.4$ the leading term is $c^2/16 = 0.0100 = A$, the same prefactor as Paper I, but the cubic term $-c^3/16 = -0.0040$ is 40% of it at $i = 1$ and the series converges only geometrically with ratio 0.4: Mechanism A is a genuinely non-perturbative test of the $\alpha_c = \frac{1}{2}$ law, and the odd terms will be seen to matter in Section 4.

Definition 2.3 (Mechanism B: iso-spectral Givens mode rotation). Let $\Lambda_A = \Lambda_B = \Lambda$ (every amplitude unchanged), and group the modes into adjacent pairs $(2k-1, 2k)$, $k = 1, \dots, N$ (dimension $2N$), with anisotropy ratio $r = \lambda_{2k-1}/\lambda_{2k} > 1$. Let $C_A = \Lambda$ (block diagonal with blocks $\text{diag}(\lambda_{2k-1}, \lambda_{2k})$) and let C_B be obtained by rotating the k -th block by $R(\theta_k) \in \text{SO}(2)$,

$$C_B^{(k)} = R(\theta_k) \text{diag}(\lambda_{2k-1}, \lambda_{2k}) R(\theta_k)^\top, \quad \theta_k = c_\theta k^{-\alpha_\theta}. \quad (2.4)$$

Then $D_{\text{comm}} = 0$ identically and $D_{\text{cov}}(N) = D_{\text{rot}}(N) = \sum_{k \leq N} d_{\text{rot},k}$. Here N counts rotated pairs.

Proposition 2.4 (Exact per-pair contribution of Mechanism B). *For every θ ,*

$$d_{\text{rot}}(\theta) = \frac{1}{2} \ln[1 + \kappa(r) \sin^2 \theta], \quad \kappa(r) = \frac{(r-1)^2}{4r}, \quad (2.5)$$

an even function of θ , with

$$d_{\text{rot}}(\theta) = \frac{\kappa}{2} \theta^2 - \frac{\kappa(2+3\kappa)}{12} \theta^4 + O(\theta^6). \quad (2.6)$$

For $r = 3$, $\kappa = \frac{1}{3}$ and $d_{\text{rot}}(\theta) = \theta^2/6 - \theta^4/12 + O(\theta^6)$.

Proof. Write the block as $C_A = \text{diag}(a, b)$, $a = \lambda_{2k-1}$, $b = \lambda_{2k}$, and $C_B = RC_A R^\top$ with $R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$; abbreviate $C = \cos \theta$, $S = \sin \theta$. Then $C_B = \begin{pmatrix} aC^2+bS^2 & (a-b)CS \\ (a-b)CS & aS^2+bC^2 \end{pmatrix}$ and the block of $\frac{1}{2}(C_A + C_B)$ has determinant

$$\begin{aligned} \frac{1}{4}[(a + aC^2 + bS^2)(b + aS^2 + bC^2) - (a-b)^2 C^2 S^2] &= \frac{1}{4}[(a^2 + b^2)S^2 + 2ab(1 + C^2)] \\ &= ab \left[1 + \frac{(a-b)^2}{4ab} S^2 \right], \end{aligned}$$

where the middle equality expands the product, uses $C^2 + S^2 = 1$ and cancels the $(a-b)^2 C^2 S^2$ terms, and the last uses $1 + C^2 = 2 - S^2$. Since $\det C_A = \det C_B = ab$, $d_{\text{rot}} = \frac{1}{2} \ln \det \frac{C_A + C_B}{2} - \frac{1}{4} \ln(ab)^2 = \frac{1}{2} \ln[1 + \frac{(a-b)^2}{4ab} S^2]$, which is (2.5) with $\kappa = (a-b)^2/(4ab) = (r-1)^2/(4r)$. Evenness is that of $\sin^2$. For the expansion, $\sin^2 \theta = \theta^2 - \theta^4/3 + O(\theta^6)$ and $\frac{1}{2} \ln(1+u) = \frac{u}{2} - \frac{u^2}{4} + O(u^3)$ with $u = \kappa \sin^2 \theta$ give $\frac{\kappa}{2} \theta^2 - \frac{\kappa}{6} \theta^4 - \frac{\kappa^2}{4} \theta^4 + O(\theta^6)$, and $\frac{\kappa}{6} + \frac{\kappa^2}{4} = \frac{\kappa(2+3\kappa)}{12}$. $\square$

Remark 2.5 (Normalisation and the geometry of $\mathcal{P}_N$). With $r = 3$ and $c_\theta^2 = 0.06$ the leading coefficient is $\kappa c_\theta^2/2 = 0.0100 = A$, and the fourth-order coefficient is $-\kappa(2+3\kappa)c_\theta^4/12 = -3.0 \times 10^{-4}$, three orders of magnitude below the leading one: Mechanism B is perturbative where Mechanism A is not. The two mechanisms are the two factors of the polar decomposition of a change of covariance on $\mathcal{P}_N = \text{GL}_N(\mathbb{R})/\text{O}(N)$ [3, 14]: in the common eigenbasis the affine-invariant geodesic from C_A to C_B is $\Lambda_A^{1/2} \exp(t \text{diag}(\eta_i)) \Lambda_A^{1/2}$, with $\eta_i = \ln(\lambda_i^{(B)}/\lambda_i^{(A)})$ the geodesic coordinates, while a rotation of the eigenbasis moves along the orbit of $\text{O}(N)$, whose geodesic coordinate is the angle $\theta \in \mathfrak{so}(2)$. Mechanism A-geo and Mechanism B are perturbations in geodesic coordinates; Mechanism A is a perturbation in the Euclidean coordinate $\delta = e^\eta - 1$. Section 4 shows that this distinction is exactly what the finite-size exponents see.

3. SPECTRAL FELDMAN–HÁJEK DICHOTOMY AND THREE-REGIME EULER–MACLAURIN ASYMPTOTICS

We use the Euler–Maclaurin expansion of the generalised harmonic numbers in the form proved in Paper I: for $s > 0$, $s \neq 1$,

$$\begin{aligned} H_N(s) &= \frac{N^{1-s}}{1-s} + \zeta(s) + \frac{N^{-s}}{2} - \frac{sN^{-s-1}}{12} + O(N^{-s-3}), \\ H_N(1) &= \ln N + \gamma_{\text{EM}} + \frac{1}{2N} - \frac{1}{12N^2} + O(N^{-4}), \end{aligned} \quad (3.1)$$

where for $0 < s < 1$ the constant is the analytic continuation of ζ (negative on $(0, 1)$, with $\zeta(s) = \frac{1}{s-1} + \gamma_{\text{EM}} + O(s-1)$).

Theorem 3.1 (Unified three-regime asymptotics for mode amplitude and mode rotation). *Let $\alpha > 0$ and write the per-index contribution of either mechanism as a convergent power series in the perturbation amplitude, $d(u) = \sum_{k \in \mathcal{K}} a_k u^k$ with $u_i = c i^{-\alpha}$: for Mechanism A, $\mathcal{K} = \{2, 3, 4, \dots\}$ and a_k as in (2.2); for Mechanism A-geo, $\mathcal{K} = \{2, 4, 6, \dots\}$ with $g_2 = \frac{1}{16}$, $g_4 = -\frac{1}{384}$, $g_6 = \frac{1}{5760}$, $\dots$; for Mechanism B, $\mathcal{K} = \{2, 4, 6, \dots\}$ with $b_2 = \kappa/2$, $b_4 = -\kappa(2 + 3\kappa)/12$, $\dots$ and $c = c_\theta$. Then for every N ,*

$$D_{\text{cov}}(N) = \sum_{k \in \mathcal{K}} a_k c^k H_N(k\alpha), \quad (3.2)$$

and, with $A := a_2 c^2$ and $\alpha \notin \{1/k : k \in \mathcal{K}\}$ unless stated:

(1) **Supercritical regime**, $0 < \alpha < \frac{1}{2}$.

$$D_{\text{cov}}(N) = \sum_{k \in \mathcal{K}, k\alpha < 1} a_k c^k \frac{N^{1-k\alpha}}{1-k\alpha} + C(\alpha) + O(N^{-2\alpha}) + O(N^{1-k_0\alpha}), \quad C(\alpha) = \sum_{k \in \mathcal{K}} a_k c^k \zeta(k\alpha), \quad (3.3)$$

where $k_0 = \min\{k \in \mathcal{K} : k\alpha > 1\}$; the leading term is $\frac{A}{1-2\alpha} N^{1-2\alpha} \rightarrow \infty$. (If $k\alpha = 1$ for some k , the corresponding term is $a_k c^k (\ln N + \gamma_{\text{EM}})$.)

(2) **Critical regime**, $\alpha = \frac{1}{2}$.

$$D_{\text{cov}}(N) = A \ln N + C_{\text{crit}} + O(N^{-\beta}), \quad C_{\text{crit}} = A \gamma_{\text{EM}} + \sum_{k \in \mathcal{K}, k \geq 3} a_k c^k \zeta(k/2), \quad (3.4)$$

with $\beta = \frac{1}{2}$ for Mechanism A (from the $k = 3$ term) and $\beta = 1$ for the even mechanisms.

(3) **Subcritical regime**, $\alpha > \frac{1}{2}$.

$$\begin{aligned} D_\infty(\alpha) &= \lim_{N \rightarrow \infty} D_{\text{cov}}(N) = \sum_{k \in \mathcal{K}} a_k c^k \zeta(k\alpha) < \infty, \\ D_{\text{cov}}(N) &= D_\infty(\alpha) - \frac{A}{2\alpha - 1} N^{1-2\alpha} + O(N^{1-k_1\alpha}), \end{aligned} \quad (3.5)$$

k_1 the second element of $\mathcal{K}$.

Consequently, for each mechanism separately, the infinite-dimensional Gaussian ensembles satisfy $P_A \perp P_B$ for $\alpha \leq \frac{1}{2}$ and $P_A \sim P_B$ for $\alpha > \frac{1}{2}$: $\alpha_c = \frac{1}{2}$ is the critical exponent of the mode amplitudes and of the mode orientations alike.

Proof. For Mechanism A, $u_i = \delta_i = c i^{-\alpha} \leq c < 1$, so (2.2) converges absolutely at every i and $D_{\text{cov}}(N) = \sum_{i \leq N} \sum_k a_k \delta_i^k = \sum_k a_k c^k \sum_{i \leq N} i^{-k\alpha}$, the interchange being justified by absolute convergence ($\sum_k |a_k| c^k < \infty$); this is (3.2). For Mechanism B, $d_{\text{rot}}(\theta) = \frac{1}{2} \ln(1 + \kappa \sin^2 \theta)$ is even and analytic in θ with radius of convergence at least the distance to the nearest zero of $1 + \kappa \sin^2 \theta$, which is $|\theta| = \text{arcsinh}(\kappa^{-1/2}) > 1$ for $\kappa = \frac{1}{3}$; since $\theta_k \leq c_\theta < 1$ the same argument applies. For Mechanism A-geo, $\frac{1}{2} \ln \cosh(\eta/2)$ is even and entire in η except at $\eta = \pm i\pi$, and $\eta_i \leq c < \pi$.

Insert (3.1) into (3.2). (1) For $k\alpha < 1$ the term $N^{1-k\alpha}/(1-k\alpha)$ grows; for $k\alpha > 1$ it decays as $N^{1-k\alpha}$ (dominated by $k = k_0$); the constants assemble into $C(\alpha)$, which converges

absolutely because $|\zeta(k\alpha)|$ is bounded for $k\alpha$ away from 1 and $\sum |a_k|c^k < \infty$; the $N^{-k\alpha}/2$ terms are $O(N^{-2\alpha})$. Since all terms with $k\alpha < 1$ have $1 - k\alpha < 1 - 2\alpha$ for $k \geq 3$, the $k = 2$ term leads. (2) At $\alpha = \frac{1}{2}$ the $k = 2$ term is $A(\ln N + \gamma_{\text{EM}}) + O(N^{-1})$ and every $k \geq 3$ term converges, with tail $O(N^{1-k/2})$; the largest tail is $k = 3$ ($N^{-1/2}$) for Mechanism A and $k = 4$ (N^{-1}) for the even mechanisms. (3) For $\alpha > \frac{1}{2}$ every $k\alpha > 1$, each $H_N(k\alpha) \uparrow \zeta(k\alpha)$, and dominated convergence gives the limit; the finite- N correction is dominated by the $k = 2$ tail $-\frac{A}{2\alpha-1}N^{1-2\alpha}$, the next being $O(N^{1-k_1\alpha})$.

Singularity or equivalence: $D_{\text{cov}}(N)$ is the Bhattacharyya distance of the N -dimensional (respectively $2N$ -dimensional) projections, so the affinity of the projections is $e^{-D_{\text{cov}}(N)}$; as in Paper II the affinity of the infinite Gaussian measures is the limit of the projected affinities (martingale convergence of the density ratios, Kakutani's argument [9]), hence zero iff $D_{\text{cov}}(N) \rightarrow \infty$, and by Feldman-Hájek [7, 8] there is no third case. Here $D_{\text{cov}}(N) \rightarrow \infty$ iff $\alpha \leq \frac{1}{2}$ by (1)–(3). Equivalently, $D_{\text{comm}} \asymp \sum \eta_i^2$ and $D_{\text{rot}} \asymp \sum \theta_k^2$ (Propositions 2.2 and 2.4), so the criterion is the square-summability $\sum_i \eta_i^2 < \infty$ resp. $\sum_k \theta_k^2 < \infty$, the spectral form of the Kakutani condition. $\square$

Remark 3.2 (Numerical values of the constants). The companion script evaluates (3.2)–(3.5) with forty series terms. At $\alpha = \frac{1}{2}$: $C_{\text{crit}} = -0.0028681714$ (A), $+0.0056633399$ (A-geo), $+0.0052868221$ (B); the measured regressions of D_{cov} on $\ln N$ give slopes 0.009954, 0.009999, 0.009999 and intercepts -0.002341 , 0.005672 , 0.005296 —the even mechanisms agree with $A = 0.0100$ and C_{crit} to 10^{-5} , while Mechanism A's $N^{-1/2}$ correction from the $k = 3$ term ($+\frac{c^3}{16} \cdot 2N^{-1/2} = 0.008N^{-1/2}$ inside the window) shifts its regression slope by -5×10^{-5} and its intercept by $+5 \times 10^{-4}$. The limits are $D_{\infty}(0.75) = 0.0215672275$, $D_{\infty}(1) = 0.0127838165$ (A); 0.0260443577 , 0.0163779007 (A-geo); 0.0257702719 , 0.0161315225 (B). The doubling increments at $\alpha = \frac{1}{2}$ converge to $A \ln 2 = 0.0069315$: measured 0.0069240 (A, $N = 10^5$, the deficit being the $k = 3$ term) and 0.0069315 (B). Tables 1 and 2 compare the multi-term prediction with the measured $D_{\text{cov}}(N)$: the agreement is 3×10^{-12} or better for $N \geq 100$ and 10^{-8} at $N = 10$, where the truncation of the asymptotic expansion (3.1) after the N^{-s-1} term is the limiting factor.

4. THE DOUBLING TAIL-INCREMENT OPERATOR AND THE GEODESIC PARITY THEOREM

Theorem 4.1 (Constant-term annihilation via the doubling tail increment). *Let $D_{\text{cov}}(N) = \sum_{k \in \mathcal{K}} a_k c^k H_N(k\alpha)$ be as in Theorem 3.1 and define the doubling tail increment*

$$D_{\text{tail}}(N) := D_{\text{cov}}(2N) - D_{\text{cov}}(N) = \sum_{i=N+1}^{2N} d_i. \quad (4.1)$$

Then, for every $\alpha \in (0, 1)$ with $k\alpha \neq 1$ for all $k \in \mathcal{K}$,

$$D_{\text{tail}}(N) = \sum_{k \in \mathcal{K}} a_k c^k \frac{(2N)^{1-k\alpha} - N^{1-k\alpha}}{1 - k\alpha} + O(N^{-2\alpha}) = A \frac{2^{1-2\alpha} - 1}{1 - 2\alpha} N^{1-2\alpha} [1 + R_{\text{sub}}(N)], \quad (4.2)$$

where every constant $a_k c^k \zeta(k\alpha)$ has cancelled identically and

$$R_{\text{sub}}(N) = \frac{a_{k_1} c^{k_1}}{a_2 c^2} \frac{1 - 2\alpha}{1 - k_1\alpha} \frac{2^{1-k_1\alpha} - 1}{2^{1-2\alpha} - 1} N^{-(k_1-2)\alpha} + O(N^{-(k_2-2)\alpha}) + O(N^{-1}), \quad (4.3)$$

$k_1 < k_2$ the second and third elements of $\mathcal{K}$. Consequently the local exponent $\gamma_{\text{tail}}(\alpha; N) = d \ln D_{\text{tail}} / d \ln N$ satisfies

$$\lim_{N \rightarrow \infty} \gamma_{\text{tail}}(\alpha; N) = 1 - 2\alpha \quad \text{for all } \alpha \in (0, 1) \setminus \{\frac{1}{2}\}, \quad (4.4)$$

including the negative values $1 - 2\alpha < 0$ of the subcritical regime, while at $\alpha = \frac{1}{2}$, $D_{\text{tail}}(N) \rightarrow A \ln 2$ and $\gamma_{\text{tail}} \rightarrow 0$. Hence $\gamma_{\text{tail}}^+(\alpha) = \max(\gamma_{\text{tail}}(\alpha), 0) \rightarrow \max(1 - 2\alpha, 0)$: the doubling estimator recovers the sharp phase curve without the pole-induced rounding of the naive exponent.

Proof. Apply (3.1) at $2N$ and at N and subtract termwise in (3.2): the constants $\zeta(k\alpha)$ cancel exactly, the power terms give the first sum in (4.2), and the $\frac{1}{2}N^{-k\alpha}$ terms give $\frac{1}{2}((2N)^{-k\alpha} - N^{-k\alpha}) = O(N^{-2\alpha})$, i.e. relative $O(N^{-1})$ against the leading power. Factoring out the $k = 2$ term yields (4.3), since the ratio of the k -th to the second term is $\frac{a_k c^k}{a_2 c^2} \frac{1-2\alpha}{1-k\alpha} \frac{2^{1-k\alpha}-1}{2^{1-2\alpha}-1} N^{-(k-2)\alpha}$. All correction exponents $-(k-2)\alpha$ and -1 are negative, so $R_{\text{sub}} \rightarrow 0$ and the logarithmic derivative tends to $1-2\alpha$; the statement holds for $\alpha > \frac{1}{2}$ as well because (3.1) holds for $s > 1$ with the same form and the leading power $N^{1-2\alpha}$ is then decaying. At $\alpha = \frac{1}{2}$ the $k = 2$ term is $A(\ln 2N - \ln N) = A \ln 2$ and all others decay. $\square$

Theorem 4.2 (Refined naive-slope finite-size bias formula). *Let $\tilde{D}(N) = \sum_{k \in \mathcal{K}} a_k c^k \left[\frac{N^{1-k\alpha}}{1-k\alpha} + \zeta(k\alpha) \right]$ (with $a_k c^k (\ln N + \gamma_{\text{EM}})$ when $k\alpha = 1$) be the Euler–Maclaurin form of $D_{\text{cov}}(N)$ without the decaying $N^{-k\alpha}$ terms. Its local exponent is*

$$\gamma_{\text{naive}}^{\text{ref}}(N; \alpha) = \frac{d \ln \tilde{D}}{d \ln N} = \frac{\sum_{k \in \mathcal{K}} a_k c^k N^{1-k\alpha}}{\tilde{D}(N)}, \quad (4.5)$$

which reduces to Paper I’s formula $\frac{1-2\alpha}{1+(1-2\alpha)\zeta(2\alpha)N^{2\alpha-1}}$ when only $k = 2$ is kept. The least-squares slope of $\ln \tilde{D}$ over the regression window reproduces the measured naive exponent γ_{naive} of Mechanism A to four decimals at every $\alpha \in [0.05, 0.95]$ (Table 3); at $\alpha = 0.45$ the measured value is 0.1729, the window regression of (4.5) gives 0.1729, its value at $N_ = 10^4$ is 0.1708, and Paper I’s two-term formula gives 0.1601.*

Proof. Differentiate $\tilde{D}$ with respect to $\ln N$: $dN^{1-s}/d \ln N = (1-s)N^{1-s}$ cancels the denominator $1-s$, $d \ln N/d \ln N = 1 = N^0$ handles $k\alpha = 1$, and the constants drop, giving (4.5). With $\mathcal{K} = \{2\}$, $\tilde{D} = A[N^{1-2\alpha}/(1-2\alpha) + \zeta(2\alpha)]$ and (4.5) is the formula of [5, Thm. 4.1]. The numerical statement is read from Table 3 (columns $\gamma_{\text{naive}}^{(A)}$ and $\gamma_{\text{naive}}^{\text{ref,fit}}$), computed by the companion script with forty terms: the two columns coincide to the four decimals shown at all nineteen values of α . The difference between the window regression (0.1729) and the centre value (0.1708) is the curvature of $\ln \tilde{D}$ over two decades of N , which a least-squares line averages; the difference between the refined and the two-term formula is the $k = 3$ term, whose contribution $-\frac{c^3}{16}[N^{1-3\alpha}/(1-3\alpha) + \zeta(3\alpha)]$ is not small for $c = 0.4$. $\square$

Theorem 4.3 (Geodesic parity versus Euclidean coordinate curvature). (1) *Let $\eta = \ln(\lambda^{(B)}/\lambda^{(A)})$ be the geodesic coordinate of the mode. Then the per-mode divergence is an even function of η ,*

$$d_{\text{cov}}(\eta) = \frac{1}{2} \ln \cosh \frac{\eta}{2} = \frac{\eta^2}{16} - \frac{\eta^4}{384} + \frac{\eta^6}{5760} - \dots, \quad (4.6)$$

all odd coefficients being zero; likewise $d_{\text{rot}}(\theta)$ is even in the geodesic angle θ .

(2) *The odd terms of Mechanism A are the pull-back of (4.6) through the Euclidean coordinate $\delta = e^\eta - 1$, $\eta = \ln(1 + \delta) = \delta - \frac{\delta^2}{2} + \frac{\delta^3}{3} - \dots$:*

$$\frac{1}{16} \left(\delta - \frac{\delta^2}{2} + \frac{\delta^3}{3} \right)^2 - \frac{\delta^4}{384} = \frac{\delta^2}{16} - \frac{\delta^3}{16} + \frac{7\delta^4}{128} + O(\delta^5), \quad (4.7)$$

in agreement with (2.3).

(3) *The leading finite-size shift of the doubling-tail exponent of Mechanism A is*

$$\Delta \gamma_{\text{tail}}^{(A)}(\alpha; N) = \gamma_{\text{tail}}^{(A)}(\alpha; N) - (1-2\alpha) = \alpha c \frac{1-2\alpha}{1-3\alpha} \frac{2^{1-3\alpha}-1}{2^{1-2\alpha}-1} N^{-\alpha} + O(N^{-2\alpha}) + O(N^{-1}), \quad (4.8)$$

positive for $0 < \alpha < 1$, whereas for Mechanism A-geo and Mechanism B the shift is $O(N^{-2\alpha})$.

(4) *At $\alpha = 0.45$, $c = 0.4$: (4.8) gives +0.00245 at $N_* = 10^4$ and +0.00292 averaged over the regression grid; the exact analytic doubling tail (4.2) regressed over the window gives $\gamma_{\text{tail}} = 0.10271$ (three terms 0.10275, two terms 0.10000); the measured values are 0.10277 (A), 0.10006 (A-geo) and 0.10006 (B).*

Proof. (1) $\frac{\lambda^{(A)} + \lambda^{(B)}}{2\sqrt{\lambda^{(A)}\lambda^{(B)}}} = \frac{1}{2}(e^{\eta/2} + e^{-\eta/2}) = \cosh(\eta/2)$, and $\ln \cosh u = \frac{u^2}{2} - \frac{u^4}{12} + \frac{u^6}{45} - \dots$ is even; with $u = \eta/2$ and the factor $\frac{1}{2}$ the coefficients are $\frac{1}{16}, -\frac{1}{384}, \frac{1}{5760}$. Evenness of d_{rot} is Proposition 2.4. (2) $\eta^2 = \delta^2 - \delta^3 + (\frac{1}{4} + \frac{2}{3})\delta^4 + O(\delta^5) = \delta^2 - \delta^3 + \frac{11}{12}\delta^4 + O(\delta^5)$; dividing by 16 and subtracting $\eta^4/384 = \delta^4/384 + O(\delta^5)$ gives $\frac{\delta^2}{16} - \frac{\delta^3}{16} + (\frac{11}{192} - \frac{1}{384})\delta^4 = \frac{\delta^2}{16} - \frac{\delta^3}{16} + \frac{21}{384}\delta^4$, and $\frac{21}{384} = \frac{7}{128}$; the companion script checks the identity $\frac{1}{2} \ln \cosh(\eta/2) = \frac{1}{2} \ln \frac{1+\delta/2}{\sqrt{1+\delta}}$ at $\delta = 0.4$ to thirty digits. (3) In (4.3) the second element of $\mathcal{K}$ is $k_1 = 3$ for Mechanism A, with $a_3 c^3 / (a_2 c^2) = -c$, so $R_{\text{sub}} = -c \frac{1-2\alpha}{1-3\alpha} \frac{2^{1-3\alpha}-1}{2^{1-2\alpha}-1} N^{-\alpha} + \dots$; the logarithmic derivative of $N^{1-2\alpha}(1 + R_{\text{sub}})$ is $(1 - 2\alpha) + dR_{\text{sub}}/d \ln N + O(R_{\text{sub}}^2) = (1 - 2\alpha) - \alpha R_{\text{sub}}$, which is (4.8). Positivity: $\frac{2^{1-3\alpha}-1}{1-3\alpha} > 0$ and $\frac{2^{1-2\alpha}-1}{1-2\alpha} > 0$ for all α (both are of the form $(2^x - 1)/x > 0$), so the sign is that of $\alpha c > 0$. For the even mechanisms $k_1 = 4$, so $R_{\text{sub}} = O(N^{-2\alpha})$. (4) The numbers are from the companion script (Table 4 and the summary file); the first-order formula evaluated at the window centre underestimates the least-squares value because $N^{-\alpha}$ is larger at the left end of the window and the higher-order terms of (4.3) are not negligible at $c = 0.4$, while the full analytic tail, regressed on the same grid as the data, matches the measured exponent to 6×10^{-5} . $\square$

Remark 4.4 (Reading the three-way comparison). Table 4 places the three mechanisms side by side. The geodesic and rotational exponents are within 10^{-4} of $1 - 2\alpha$ at every one of the nineteen values of α , including the subcritical negative values down to -0.90 ; the Euclidean exponent carries the positive shift (4.8), largest at small α where $N^{-\alpha}$ decays slowest ($+0.014$ at $\alpha = 0.10$) and $+0.0028$ at $\alpha = 0.45$. This is the answer to question (3) of Section 1: the difference is not numerical, it is the sign of the cubic coefficient $a_3 = -1/16$, and it disappears in geodesic coordinates. The practical rule for empirical spectra is therefore to fingerprint amplitudes by $\eta_i = \ln(\lambda_i^{(B)}/\lambda_i^{(A)})$, as Paper II already did, and never by the ratio $\lambda_i^{(B)}/\lambda_i^{(A)} - 1$.

5. THE TWO-DIMENSIONAL $(\alpha_\lambda, \alpha_\theta)$ JOINT PHASE DIAGRAM AND CRYPTIC ALLOSTERY

Theorem 5.1 (Joint spectral–rotational Feldman–Hájek phase diagram). *Let the k -th block ($k \geq 1$) carry both perturbations, $x_{1,k} = c(2k-1)^{-\alpha_\lambda}$, $x_{2,k} = c(2k)^{-\alpha_\lambda}$ and $\theta_k = c_\theta k^{-\alpha_\theta}$, with $r = 3$, $c = 0.4$, $c_\theta = \sqrt{0.06}$. Then*

$$P_A \sim P_B \iff (\alpha_\lambda, \alpha_\theta) \in (\frac{1}{2}, \infty) \times (\frac{1}{2}, \infty),$$

$$\gamma_{\text{eff}}(\alpha_\lambda, \alpha_\theta) := \lim_{N \rightarrow \infty} \gamma_{\text{tail}}^+ = \max(1 - 2 \min(\alpha_\lambda, \alpha_\theta), 0). \quad (5.1)$$

The $(\alpha_\lambda, \alpha_\theta)$ plane therefore splits into four phases: both parts divergent ($\alpha_\lambda, \alpha_\theta \leq \frac{1}{2}$); amplitude-driven singularity ($\alpha_\lambda \leq \frac{1}{2} < \alpha_\theta$); rotation-driven singularity ($\alpha_\theta \leq \frac{1}{2} < \alpha_\lambda$); and equivalence. On the 19×19 grid $\alpha_\lambda, \alpha_\theta \in \{0.05, 0.10, \dots, 0.95\}$ the measured γ_{tail}^+ classifies all 361 points correctly (280 singular, 81 equivalent, no mismatch), with maximal deviation 0.033 from (5.1) (Table 5).

The proof rests on an exact block formula that contains Propositions 2.2 and 2.4 as special cases.

Lemma 5.2 (Exact block decomposition when both mechanisms act). *Let a 2×2 block have $C_A = \text{diag}(a, b)$ and $C_B = R(\theta) \text{diag}(a(1+x_1), b(1+x_2)) R(\theta)^\top$, $a > b > 0$, $x_1, x_2 > -1$. Then, with $\ell_a = a(1+x_1)$, $\ell_b = b(1+x_2)$,*

$$D_{\text{cov}}^{\text{block}} = \underbrace{\frac{1}{2} \ln \frac{a + \ell_a}{2\sqrt{a\ell_a}} + \frac{1}{2} \ln \frac{b + \ell_b}{2\sqrt{b\ell_b}}}_{D_{\text{comm}}^{\text{block}}} + \underbrace{\frac{1}{2} \ln \left[1 + \frac{(a-b)(\ell_a - \ell_b)}{(a + \ell_a)(b + \ell_b)} \sin^2 \theta \right]}_{D_{\text{rot}}^{\text{block}}}. \quad (5.2)$$

For $a/b = r$ the rotational coefficient is $\kappa(r; x_1, x_2) = \frac{(r-1)(r(1+x_1)-(1+x_2))}{r(2+x_1)(2+x_2)}$, which reduces to $\kappa(r)$ at $x_1 = x_2 = 0$ and is bounded between positive constants uniformly for $x_1, x_2 \in [0, c]$ whenever $r(1+x_1) > 1+x_2$; for $r = 3$, $c = 0.4$ it lies in $[0.185, 0.533]$.

Proof. With $C = \cos \theta$, $S = \sin \theta$, $C_B = \begin{pmatrix} \ell_a C^2 + \ell_b S^2 & (\ell_a - \ell_b)CS \\ (\ell_a - \ell_b)CS & \ell_a S^2 + \ell_b C^2 \end{pmatrix}$, and $4 \det \frac{C_A + C_B}{2} = (a + \ell_a C^2 + \ell_b S^2)(b + \ell_a S^2 + \ell_b C^2) - (\ell_a - \ell_b)^2 C^2 S^2$. Expanding and using $\det C_B = \ell_a \ell_b = (\ell_a C^2 + \ell_b S^2)(\ell_a S^2 + \ell_b C^2) - (\ell_a - \ell_b)^2 C^2 S^2$,

$$\begin{aligned} 4 \det \frac{C_A + C_B}{2} &= ab + \ell_a \ell_b + a(\ell_a S^2 + \ell_b C^2) + b(\ell_a C^2 + \ell_b S^2) \\ &= (a + \ell_a)(b + \ell_b) + (a - b)(\ell_a - \ell_b)S^2, \end{aligned}$$

where the last step writes $a\ell_a S^2 + a\ell_b C^2 + b\ell_a C^2 + b\ell_b S^2 = a\ell_b + b\ell_a + (a\ell_a + b\ell_b - a\ell_b - b\ell_a)S^2$. Hence $D_{\text{cov}}^{\text{block}} = \frac{1}{2} \ln \frac{(a+\ell_a)(b+\ell_b)}{4\sqrt{ab\ell_a\ell_b}} + \frac{1}{2} \ln[1 + \frac{(a-b)(\ell_a-\ell_b)}{(a+\ell_a)(b+\ell_b)}S^2]$, and the first term is the commuting part of the ordered spectra (a, b) , (ℓ_a, ℓ_b) (Theorem 2.3 of Paper II; the ordering $\ell_a \geq \ell_b$ holds since $r(1+x_1) > 1+x_2$). Substituting $a = rb$, $\ell_a = rb(1+x_1)$, $\ell_b = b(1+x_2)$ gives $\kappa(r; x_1, x_2)$; for $r = 3$ and $x_1, x_2 \in [0, 0.4]$ the numerator $2(3(1+x_1) - (1+x_2))$ lies in $[3.2, 6.4]$ and the denominator $3(2+x_1)(2+x_2)$ in $[12, 17.28]$. $\square$

Proof of Theorem 5.1. By Lemma 5.2, $D_{\text{cov}}(N) = D_{\text{comm}}(N) + D_{\text{rot}}(N)$ with both parts non-negative: $D_{\text{comm}}(N) = \sum_{k \leq N} [d_{\text{cov}}(x_{1,k}) + d_{\text{cov}}(x_{2,k})]$ and $D_{\text{rot}}(N) = \sum_{k \leq N} \frac{1}{2} \ln[1 + \kappa_k \sin^2 \theta_k]$ with $\kappa_k \in [0.185, 0.533]$. Since $d_{\text{cov}}(x) \asymp x^2$ on $[0, c]$ (Proposition 2.2: $d_{\text{cov}}(x)/x^2 \in [a_2 - |a_3|c - \dots, a_2]$ is bounded between positive constants) and $\frac{1}{2} \ln[1 + \kappa_k \sin^2 \theta] \asymp \theta^2$ on $[0, c_\theta]$ uniformly in k , $D_{\text{comm}}(N)$ is bounded iff $\sum_k k^{-2\alpha_\lambda} < \infty$ iff $\alpha_\lambda > \frac{1}{2}$, and $D_{\text{rot}}(N)$ is bounded iff $\sum_k k^{-2\alpha_\theta} < \infty$ iff $\alpha_\theta > \frac{1}{2}$. $D_{\text{cov}}(N)$ is bounded iff both are, which with the Feldman-Hájek argument of Theorem 3.1 gives the equivalence criterion. For the exponent, Theorem 4.1 applied to each part gives $D_{\text{tail}}^{\text{comm}} \asymp N^{1-2\alpha_\lambda}$ and $D_{\text{tail}}^{\text{rot}} \asymp N^{1-2\alpha_\theta}$ (up to the sub-leading corrections of (4.3) and, for the rotation part, the slowly varying factor $\kappa_k \rightarrow \kappa(r)$), so the doubling tail of the sum is governed by the larger exponent and $\gamma_{\text{tail}} \rightarrow \max(1-2\alpha_\lambda, 1-2\alpha_\theta)$, whence (5.1). The grid statement is the content of `exp03_results_phase_diagram.csv`: a point is classified singular when its measured $\gamma_{\text{tail}} > -0.05$ (i.e. D_{tail} does not decay), which agrees with $\min(\alpha_\lambda, \alpha_\theta) \leq \frac{1}{2}$ at all 361 points. The finite- N deviations of γ_{tail}^+ from (5.1) are the cross-over of two power laws inside the window; Corollary 5.3 gives them in closed form. $\square$

Corollary 5.3 (Finite- N two-power-law cross-over formula across all singular quadrants). *Let $\gamma_\lambda = 1 - 2\alpha_\lambda$, $\gamma_\theta = 1 - 2\alpha_\theta \in (-1, 1)$ and let the block model of Theorem 5.1 be given.*

- (1) *The joint doubling tail increment is the sum of two power-law sequences,*

$$D_{\text{tail}}(N) = D_{\text{tail}}^{\text{comm}}(N) + D_{\text{tail}}^{\text{rot}}(N) = W_\lambda N^{\gamma_\lambda} [1 + R_{\text{sub}}^{\text{comm}}(N)] + W_\theta N^{\gamma_\theta} [1 + R_{\text{sub}}^{\text{rot}}(N)], \quad (5.3)$$

with the weights

$$W_\lambda = 2^{\gamma_\lambda} A \frac{2^{\gamma_\lambda} - 1}{\gamma_\lambda}, \quad W_\theta = \bar{\kappa} A \frac{2^{\gamma_\theta} - 1}{\gamma_\theta}, \quad \bar{\kappa} = \kappa_k / \kappa(r) \rightarrow 1, \quad (5.4)$$

(with $(2^\gamma - 1)/\gamma$ read as $\ln 2$ at $\gamma = 0$): the factor 2^{γ_λ} arises because the k -th block carries the two modes $2k-1, 2k$, whose amplitude contributions sum to $2^{1-2\alpha_\lambda} A k^{-2\alpha_\lambda} (1+o(1))$, while it carries a single rotation angle.

- (2) *The local exponent of the sum is the convex combination of the two local exponents weighted by the two increments,*

$$\gamma_{\text{tail}}^{\text{joint}}(N) = \frac{\gamma_{\text{tail}}^{\text{comm}}(N) D_{\text{tail}}^{\text{comm}}(N) + \gamma_{\text{tail}}^{\text{rot}}(N) D_{\text{tail}}^{\text{rot}}(N)}{D_{\text{tail}}^{\text{comm}}(N) + D_{\text{tail}}^{\text{rot}}(N)} = \gamma_1 - \frac{\gamma_1 - \gamma_2}{1 + \frac{W_1}{W_2} N^{\gamma_1 - \gamma_2}} + \Delta\gamma_{\text{sub}}(N), \quad (5.5)$$

where $\gamma_1 = \max(\gamma_\lambda, \gamma_\theta)$, $\gamma_2 = \min(\gamma_\lambda, \gamma_\theta)$, W_1, W_2 are the corresponding weights, and $\Delta\gamma_{\text{sub}}$ collects the sub-leading corrections of Theorem 4.1, in particular the Euclidean shift (4.8) of the amplitude part. The subtracted term is the pull of the subdominant power law, which can be a decaying one ($\gamma_2 < 0$) in the single-divergence quadrants.

- (3) *On the 19×19 grid the leading closed form (the first two terms of (5.5) at $N_* = 10^4$) reproduces the measured γ_{tail} to 0.013, and the full analytic prediction (all series terms of both parts, regressed over the window as the data are) to 4×10^{-4} at every one of the 361 points (root-mean-square 8×10^{-5}). The four characteristic entries of Table 5 are explained in the proof.*

$(\alpha_\lambda, \alpha_\theta)$	γ_λ	γ_θ	W_λ	W_θ	leading form at N_*	full analytic fit	measured γ_{tail}
(0.55, 0.45)	-0.10	+0.10	0.00625	0.00718	+0.0758	+0.0756	+0.0756
(0.45, 0.55)	+0.10	-0.10	0.00769	0.00670	+0.0758	+0.0770	+0.0771
(0.35, 0.45)	+0.30	+0.10	0.00949	0.00718	+0.2786	+0.2816	+0.2817
(0.45, 0.35)	+0.10	+0.30	0.00769	0.00771	+0.2727	+0.2727	+0.2728
(0.45, 0.45)	+0.10	+0.10	0.00769	0.00718	+0.1000	+0.1010	+0.1011
(0.25, 0.25)	+0.50	+0.50	0.01172	0.00828	+0.5000	+0.5045	+0.5046

Proof. By Lemma 5.2, $D_{\text{tail}} = D_{\text{tail}}^{\text{comm}} + D_{\text{tail}}^{\text{rot}}$ with $D_{\text{tail}}^{\text{comm}}(N) = \sum_{k=N+1}^{2N} [d_{\text{cov}}(x_{1,k}) + d_{\text{cov}}(x_{2,k})]$. The modes of the blocks $N+1, \dots, 2N$ are the modes $2N+1, \dots, 4N$, so $D_{\text{tail}}^{\text{comm}}(N)$ is the doubling tail of Theorem 4.1 evaluated at $2N$: by (4.2), $D_{\text{tail}}^{\text{comm}}(N) = A^{\frac{2\gamma_\lambda-1}{\gamma_\lambda}} (2N)^{\gamma_\lambda} [1 + R_{\text{sub}}^{\text{comm}}]$, which is the first term of (5.3) with $W_\lambda = 2^{\gamma_\lambda} A(2^{\gamma_\lambda} - 1)/\gamma_\lambda$. For the rotation part, $\frac{1}{2} \ln[1 + \kappa_k \sin^2 \theta_k] = \frac{\kappa_k}{2} \theta_k^2 (1 + O(\theta_k^2))$ with $\kappa_k = \kappa(r; x_{1,k}, x_{2,k}) = \kappa(r)(1 + O(x_{1,k} + x_{2,k}))$ by Lemma 5.2, so the pure-rotation tail of Theorem 4.1 applies with the slowly varying factor $\bar{\kappa} = \kappa_k/\kappa(r) \rightarrow 1$ absorbed into the sub-leading remainder: this is the second term with $W_\theta = \bar{\kappa} A(2^{\gamma_\theta} - 1)/\gamma_\theta$. For the exponent, $\frac{d}{d \ln N} \ln(D_1 + D_2) = \frac{\gamma_1 D_1 + \gamma_2 D_2}{D_1 + D_2}$ with $\gamma_j = d \ln D_j / d \ln N$, which is the first form of (5.5); inserting $D_j = W_j N^{\gamma_j}$ and writing $\frac{\gamma_1 D_1 + \gamma_2 D_2}{D_1 + D_2} = \gamma_1 - (\gamma_1 - \gamma_2) \frac{D_2}{D_1 + D_2} = \gamma_1 - \frac{\gamma_1 - \gamma_2}{1 + D_1/D_2}$ gives the closed form, the remainders R_{sub} contributing $\Delta\gamma_{\text{sub}}$. The grid statement is computed by the companion script (file `results/paper03_table6_crossover.csv`): the leading form deviates from the measured exponents by at most 0.013 (both-divergent and amplitude quadrants, where the Euclidean shift of the amplitude part is largest) and by at most 0.0012 in the rotation quadrant; the full analytic prediction deviates by at most 0.0004 over all 361 points.

The four entries. At (0.55, 0.45) the rotation part dominates with $\gamma_1 = +0.10$ and the amplitude part is subcritical, $\gamma_2 = -0.10$, a *decaying* tail of positive weight: $W_\theta/W_\lambda = 1.149$, $N_*^{0.20} = 6.31$, so the pull is $0.20/(1 + 1.149 \times 6.31) = 0.0242$ and the local exponent is $0.100 - 0.024 = 0.076$, the measured value; the mirror point (0.45, 0.55) has the same leading form 0.0758 but the dominant part is now the Euclidean amplitude part, whose shift (4.8) adds +0.0013 after weighting, giving 0.0770 against the measured 0.0771. These are the maximal deviations 0.024 and 0.023 of the two single-divergence quadrants in Table 5. In the both-divergent quadrant, (0.35, 0.45) has the amplitude part dominant ($\gamma_1 = 0.30$, $W_\lambda/W_\theta = 1.322$ because of the factor $2^{0.30}$) with pull $0.20/(1 + 1.322 \times 6.31) = 0.0214$, hence 0.2786, and the Euclidean shift of the dominant part raises it to 0.2816 against the measured 0.2817; at the mirror point (0.45, 0.35) the rotation part dominates with $W_\theta/W_\lambda = 1.002$, pull 0.0273, exponent 0.2727, and the Euclidean shift now acts only on the subdominant part (+0.0001): 0.2727 against the measured 0.2728. The mirror asymmetry 0.282 versus 0.273 of Table 5 is therefore the product of the two-mode factor 2^{γ_λ} in W_λ and of the sign of a_3 ; the maximal deviation 0.033 of that quadrant occurs on the diagonal (0.25, 0.25), where the two power laws have the same exponent and the deviation is the Euclidean shift alone, $0.5046 - 0.5000$, weighted by the amplitude share $W_\lambda/(W_\lambda + W_\theta) = 0.586$ and evaluated over the window. $\square$

Remark 5.4 (Cryptic iso-spectral allostery). The rotation-driven phase is the one that no first-order observable can see. In it every eigenvalue of the covariance is unchanged—so every residue fluctuation amplitude, every principal-component variance and every entropy-like functional of the spectrum is unchanged—and yet the infinite ensembles are mutually singular, because the directions of the collective modes have been rotated by angles that decay too slowly along the spectrum. Paper II exhibited a single such pair (its Type II, with 71.6% of D_{cov} in the rotation part); Theorem 5.1 shows that the phenomenon has a critical exponent, $\alpha_\theta^c = \frac{1}{2}$,

and that it is invisible to the commuting fingerprint $\Delta\lambda_i$ alone. The rotation fingerprint θ_k must be measured, and its decay along the mode index is the quantity that decides between a localised and a global allosteric response. Figure 1D shows the four phases as measured, with the phase boundaries at $\alpha = \frac{1}{2}$ on both axes.

NUMERICAL VERIFICATION TABLES AND FIGURE

Protocol. `exp03_spectral_kakutani_criticality.py` evaluates the exact per-mode contributions (2.2) and (2.5) for $i \leq 2 \times 10^5$, forms $D_{\text{cov}}(N)$ by cumulative summation, $D_{\text{tail}}(N)$ by (4.1) and $\Pi_N^{(2)} = e^{-D_{\text{cov}}(N)}$; regresses $\ln D_{\text{cov}}$ and $\ln D_{\text{tail}}$ on $\ln N$ over the sixty-point grid in $[10^3, 10^5]$; and evaluates the joint block model of Lemma 5.2 on the 19×19 grid. `paper03_compute_tables.py` evaluates the forty-term Euler–Maclaurin predictions of Theorem 3.1, the refined slope of Theorem 4.2 (both at N_* and regressed over the window), the geodesic variant of Mechanism A and the shift formula of Theorem 4.3, the cross-over predictions of Corollary 5.3 on the full grid, and writes `results/paper03_tables_summary.txt`. All numbers in Tables 1–5 are copied from `data/exp03_results_*.csv` and `results/paper03_table4.csv`.

TABLE 1. Mechanism A (Euclidean amplitude perturbation, $c = 0.4$): measured $D_{\text{cov}}(N)$ against the forty-term Euler–Maclaurin prediction of Theorem 3.1, the doubling tail increment, the second-order affinity and, for $\alpha > \frac{1}{2}$, the limit $D_\infty(\alpha)$ of (3.5). Source: `exp03_results_benchmark.csv`.

α_λ	N	$D_{\text{cov}}(N)$	EM prediction	difference	$D_{\text{tail}}(N)$	$\Pi_N^{(2)}$	D_∞
0.25	10^1	0.0383866	0.0383866	5.5×10^{-9}	0.0212969	0.962341	–
0.25	10^2	0.1549782	0.1549782	2.1×10^{-12}	0.0740284	0.856434	–
0.25	10^3	0.5507391	0.5507391	5.6×10^{-16}	0.245825	0.576524	–
0.25	10^4	1.8512397	1.8512397	-6.2×10^{-15}	0.799081	0.157042	–
0.25	10^5	6.0543795	6.0543795	-2.5×10^{-14}	2.56683	0.002348	–
0.50	10^1	0.0229930	0.0229930	6.4×10^{-9}	0.00604584	0.977269	–
0.50	10^2	0.0440178	0.0440178	7.6×10^{-13}	0.00668026	0.956937	–
0.50	10^3	0.0664659	0.0664659	2.4×10^{-16}	0.00685561	0.935695	–
0.50	10^4	0.0893156	0.0893156	5.0×10^{-16}	0.00690786	0.914557	–
0.50	10^5	0.1122864	0.1122864	-1.9×10^{-16}	0.00692404	0.893788	–
0.75	10^1	0.0155599	0.0155599	4.9×10^{-9}	0.00166213	0.984561	0.0215672
0.75	10^2	0.0195822	0.0195822	1.8×10^{-13}	0.000576801	0.980608	0.0215672
0.75	10^3	0.0209355	0.0209355	3.5×10^{-17}	0.000184811	0.979282	0.0215672
0.75	10^4	0.0213673	0.0213673	5.6×10^{-17}	5.85569×10^{-5}	0.978859	0.0215672
0.75	10^5	0.0215040	0.0215040	-2.2×10^{-16}	1.85230×10^{-5}	0.978726	0.0215672
1.00	10^1	0.0118499	0.0118499	3.0×10^{-9}	0.00045095	0.988220	0.0127838
1.00	10^2	0.0126845	0.0126845	3.3×10^{-14}	4.94786×10^{-5}	0.987396	0.0127838
1.00	10^3	0.0127738	0.0127738	6.9×10^{-18}	4.99475×10^{-6}	0.987307	0.0127838
1.00	10^4	0.0127828	0.0127828	-5.2×10^{-17}	4.99948×10^{-7}	0.987299	0.0127838
1.00	10^5	0.0127837	0.0127837	-1.0×10^{-16}	4.99995×10^{-8}	0.987298	0.0127838

TABLE 2. Mechanism B (iso-spectral Givens rotation, $r = 3$, $c_\theta = \sqrt{0.06}$; N counts rotated pairs): the same quantities as Table 1. Source: `exp03_results_benchmark.csv`.

α_θ	N	$D_{\text{cov}}(N)$	EM prediction	difference	$D_{\text{tail}}(N)$	$\Pi_N^{(2)}$	D_∞
0.25	10^1	0.0493449	0.0493448	8.0×10^{-9}	0.0255431	0.951853	–
0.25	10^2	0.1843563	0.1843563	2.6×10^{-12}	0.0824897	0.831639	–
0.25	10^3	0.6157818	0.6157818	1.4×10^{-15}	0.261718	0.540218	–
0.25	10^4	1.9825279	1.9825279	-1.3×10^{-15}	0.828205	0.137721	–
0.25	10^5	6.3063583	6.3063583	-1.2×10^{-13}	2.6195	0.001825	–
0.50	10^1	0.0288329	0.0288329	8.2×10^{-9}	0.00667382	0.971579	–
0.50	10^2	0.0513914	0.0513914	8.3×10^{-13}	0.00690505	0.949907	–
0.50	10^3	0.0743697	0.0743697	1.1×10^{-16}	0.00692882	0.928328	–
0.50	10^4	0.0973908	0.0973908	-4.2×10^{-17}	0.00693121	0.907201	–
0.50	10^5	0.1204161	0.1204161	1.2×10^{-15}	0.00693145	0.886551	–
0.75	10^1	0.0196012	0.0196012	5.7×10^{-9}	0.00175246	0.980590	0.0257703
0.75	10^2	0.0237753	0.0237753	1.8×10^{-13}	0.000582553	0.976505	0.0257703
0.75	10^3	0.0251380	0.0251380	-5.2×10^{-17}	0.00018514	0.975175	0.0257703
0.75	10^4	0.0255703	0.0255703	-9.0×10^{-17}	5.85754×10^{-5}	0.974754	0.0257703
0.75	10^5	0.0257070	0.0257070	2.7×10^{-16}	1.85241×10^{-5}	0.974621	0.0257703
1.00	10^1	0.0151799	0.0151799	3.3×10^{-9}	0.000463881	0.984935	0.0161315
1.00	10^2	0.0160320	0.0160320	3.3×10^{-14}	4.96264×10^{-5}	0.984096	0.0161315
1.00	10^3	0.0161215	0.0161215	-2.4×10^{-17}	4.99625×10^{-6}	0.984008	0.0161315
1.00	10^4	0.0161305	0.0161305	-2.1×10^{-17}	4.99963×10^{-7}	0.983999	0.0161315
1.00	10^5	0.0161314	0.0161314	-3.9×10^{-16}	4.99996×10^{-8}	0.983998	0.0161315

TABLE 3. Scaling exponents over the regression window $[10^3, 10^5]$ (Theorems 4.1 and 4.2): asymptotic exponent $1-2\alpha$; naive exponent of Mechanism A; Paper I’s two-term prediction at N_* ; the refined prediction (4.5) at N_* and regressed over the window; doubling-tail exponent of Mechanism A and its analytic prediction from (4.2); naive and doubling-tail exponents of Mechanism B. Source: `exp03_results_scaling_comparison.csv`.

α	$1-2\alpha$	$\gamma_{\text{naive}}^{(A)}$	Paper I	ref. N_*	ref. fit	$\gamma_{\text{tail}}^{(A)}$	pred.	$\gamma_{\text{naive}}^{(B)}$	$\gamma_{\text{tail}}^{(B)}$
0.05	+0.90	0.9113	0.9001	0.9090	0.9113	+0.9105	+0.9087	0.9014	+0.9012
0.10	+0.80	0.8163	0.8003	0.8147	0.8163	+0.8138	+0.8129	0.8016	+0.8009
0.15	+0.70	0.7181	0.7007	0.7170	0.7181	+0.7135	+0.7131	0.7018	+0.7006
0.20	+0.60	0.6188	0.6016	0.6177	0.6188	+0.6115	+0.6114	0.6027	+0.6003
0.25	+0.50	0.5201	0.5037	0.5189	0.5201	+0.5092	+0.5091	0.5048	+0.5002
0.30	+0.40	0.4237	0.4080	0.4222	0.4237	+0.4070	+0.4070	0.4093	+0.4001
0.35	+0.30	0.3316	0.3167	0.3299	0.3316	+0.3052	+0.3052	0.3182	+0.3001
0.40	+0.20	0.2469	0.2327	0.2449	0.2469	+0.2038	+0.2038	0.2345	+0.2001
0.45	+0.10	0.1729	0.1601	0.1708	0.1729	+0.1028	+0.1027	0.1619	+0.1001
0.50	+0.00	0.1130	0.1022	0.1109	0.1130	+0.0020	+0.0019	0.1039	+0.0001
0.55	-0.10	0.0687	0.0603	0.0667	0.0687	-0.0986	-0.0986	0.0619	-0.0999
0.60	-0.20	0.0391	0.0330	0.0373	0.0391	-0.1990	-0.1990	0.0345	-0.1999
0.65	-0.30	0.0210	0.0170	0.0195	0.0210	-0.2993	-0.2993	0.0182	-0.2999
0.70	-0.40	0.0108	0.0083	0.0097	0.0108	-0.3995	-0.3995	0.0092	-0.3999
0.75	-0.50	0.0054	0.0039	0.0046	0.0054	-0.4996	-0.4997	0.0045	-0.4999
0.80	-0.60	0.0026	0.0017	0.0021	0.0026	-0.5997	-0.5998	0.0022	-0.5999
0.85	-0.70	0.0013	0.0008	0.0009	0.0013	-0.6997	-0.6998	0.0010	-0.6999
0.90	-0.80	0.0006	0.0003	0.0004	0.0006	-0.7998	-0.7999	0.0005	-0.7999
0.95	-0.90	0.0003	0.0001	0.0002	0.0003	-0.8998	-0.8999	0.0002	-0.8999

TABLE 4. Three-way comparison of the doubling-tail exponent (Theorem 4.3): Euclidean Mechanism A with its bias and the first-order shift (4.8) at $N_* = 10^4$; geodesic Mechanism A-geo ($\lambda_i^{(B)} = \lambda_i^{(A)} e^{0.4i^{-\alpha}}$); rotational Mechanism B. Bias = $\gamma_{\text{tail}} - (1 - 2\alpha)$. Source: `paper03_table4.csv`.

α	$1-2\alpha$	$\gamma_{\text{tail}}^{(A)}$	bias A	(4.8) at N_*	$\gamma_{\text{tail}}^{(A\text{-geo})}$	bias A-geo	$\gamma_{\text{tail}}^{(B)}$	bias B
0.05	+0.90	+0.9105	+0.0105	+0.0124	+0.9003	+0.0003	+0.9012	+0.0012
0.10	+0.80	+0.8138	+0.0138	+0.0153	+0.8002	+0.0002	+0.8009	+0.0009
0.15	+0.70	+0.7135	+0.0135	+0.0143	+0.7001	+0.0001	+0.7006	+0.0006
0.20	+0.60	+0.6115	+0.0115	+0.0118	+0.6001	+0.0001	+0.6003	+0.0003
0.25	+0.50	+0.5092	+0.0092	+0.0091	+0.5001	+0.0001	+0.5002	+0.0002
0.30	+0.40	+0.4070	+0.0070	+0.0068	+0.4001	+0.0001	+0.4001	+0.0001
0.35	+0.30	+0.3052	+0.0052	+0.0049	+0.3000	+0.0000	+0.3001	+0.0001
0.40	+0.20	+0.2038	+0.0038	+0.0035	+0.2001	+0.0001	+0.2001	+0.0001
0.45	+0.10	+0.1028	+0.0028	+0.0024	+0.1001	+0.0001	+0.1001	+0.0001
0.50	+0.00	+0.0020	+0.0020	+0.0017	+0.0001	+0.0001	+0.0001	+0.0001
0.55	-0.10	-0.0986	+0.0014	+0.0012	-0.0999	+0.0001	-0.0999	+0.0001
0.60	-0.20	-0.1990	+0.0010	+0.0008	-0.1999	+0.0001	-0.1999	+0.0001
0.65	-0.30	-0.2993	+0.0007	+0.0005	-0.2999	+0.0001	-0.2999	+0.0001
0.70	-0.40	-0.3995	+0.0005	+0.0004	-0.3999	+0.0001	-0.3999	+0.0001
0.75	-0.50	-0.4996	+0.0004	+0.0002	-0.4999	+0.0001	-0.4999	+0.0001
0.80	-0.60	-0.5997	+0.0003	+0.0002	-0.5999	+0.0001	-0.5999	+0.0001
0.85	-0.70	-0.6997	+0.0003	+0.0001	-0.6999	+0.0001	-0.6999	+0.0001
0.90	-0.80	-0.7998	+0.0002	+0.0001	-0.7999	+0.0001	-0.7999	+0.0001
0.95	-0.90	-0.8998	+0.0002	+0.0000	-0.8999	+0.0001	-0.8999	+0.0001

TABLE 5. Joint phase diagram (Theorem 5.1): measured γ_{tail}^+ on the sub-grid $\alpha_\lambda, \alpha_\theta \in \{0.05, 0.15, \dots, 0.95\}$ of the 19×19 grid (rows: α_θ , columns: α_λ). Theory: $\max(1 - 2\alpha_\lambda, 1 - 2\alpha_\theta, 0)$, i.e. 0.9, 0.7, 0.5, 0.3, 0.1 along the first five columns and rows and 0 in the lower-right block. Over the full grid: 361 points, 280 singular, 81 equivalent, 0 misclassified; maximal deviation 0.033 (both-divergent quadrant), 0.023 (amplitude quadrant), 0.024 (rotation quadrant), 0.000 (equivalent quadrant), all accounted for by the cross-over formula of Corollary 5.3. Source: `exp03_results_phase_diagram.csv`.

$\alpha_\theta \backslash \alpha_\lambda$	0.05	0.15	0.25	0.35	0.45	0.55	0.65	0.75	0.85	0.95
0.95	0.910	0.712	0.508	0.304	0.102	0.000	0.000	0.000	0.000	0.000
0.85	0.910	0.712	0.508	0.304	0.102	0.000	0.000	0.000	0.000	0.000
0.75	0.910	0.712	0.508	0.304	0.100	0.000	0.000	0.000	0.000	0.000
0.65	0.910	0.712	0.507	0.302	0.093	0.000	0.000	0.000	0.000	0.000
0.55	0.910	0.712	0.506	0.296	0.077	0.000	0.000	0.000	0.000	0.000
0.45	0.910	0.711	0.501	0.282	0.101	0.076	0.092	0.098	0.100	0.100
0.35	0.909	0.706	0.487	0.302	0.273	0.291	0.298	0.300	0.300	0.300
0.25	0.904	0.693	0.505	0.470	0.490	0.498	0.500	0.500	0.500	0.500
0.15	0.891	0.708	0.668	0.690	0.698	0.700	0.700	0.701	0.701	0.701
0.05	0.907	0.867	0.889	0.898	0.901	0.901	0.901	0.901	0.901	0.901

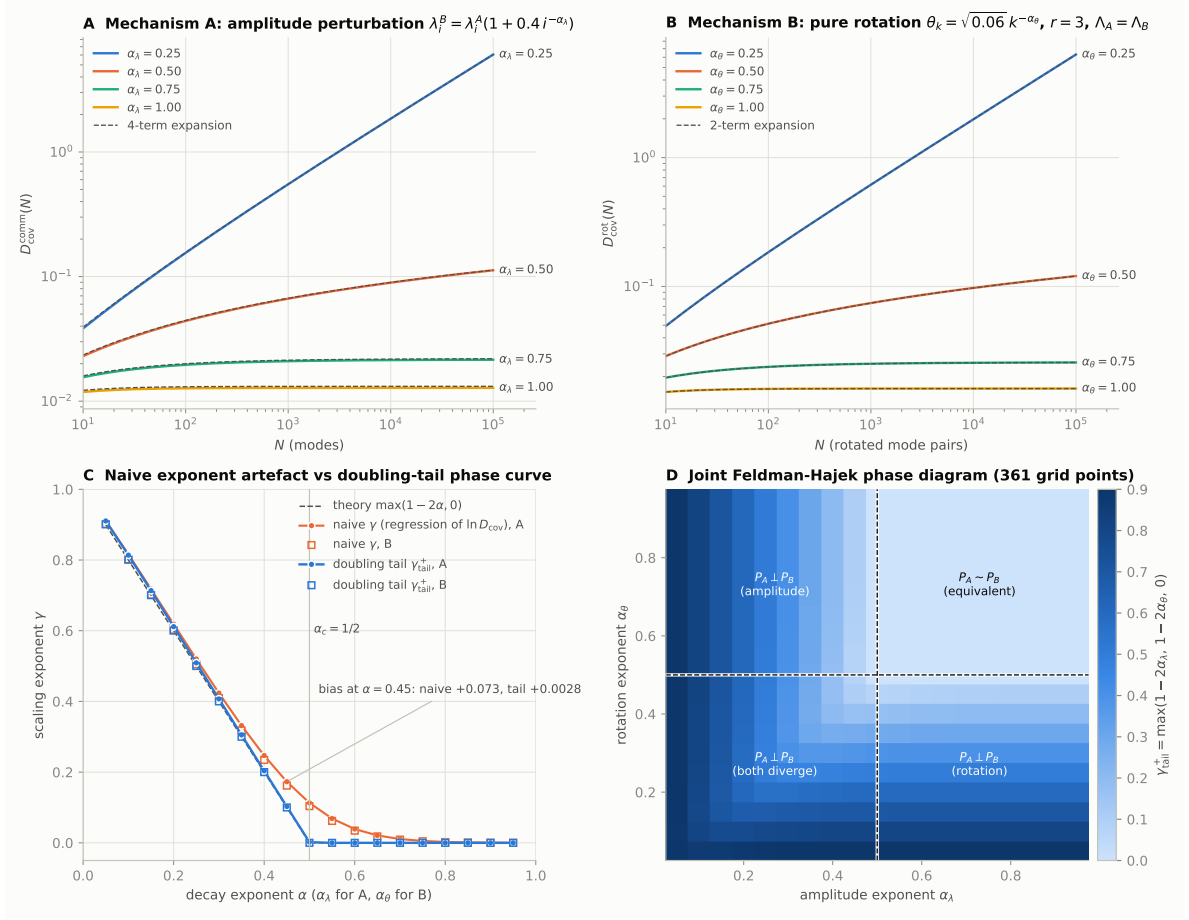


FIGURE 1. Spectral Kakutani criticality of covariance perturbations decaying along the mode index (file `fig03` in `figures/`). **A.** Mechanism A: $D_{comm}(N)$ on doubly logarithmic axes for $\alpha_\lambda \in \{0.25, 0.5, 0.75, 1\}$ with the four-term expansion (2.3) summed (dashed) superposed: power-law divergence, logarithmic divergence, saturation at $D_\infty(0.75) = 0.02157$ and $D_\infty(1) = 0.01278$. **B.** Mechanism B: $D_{rot}(N)$ for the same exponents with the two-term expansion (2.6): the same three regimes, with saturation at 0.02577 and 0.01613, although every eigenvalue is unchanged. **C.** The naive exponent (orange) is a smooth curve through $\alpha_c = 0.1729$ (A) and 0.1619 (B) at $\alpha = 0.45$, 0.113 and 0.104 at $\alpha = \frac{1}{2}$ —while the doubling-tail exponent γ_{tail}^+ (blue) is the sharp kink $\max(1 - 2\alpha, 0)$ of Theorem 4.1; filled circles are Mechanism A, open squares Mechanism B, and at $\alpha = 0.45$ the two estimators differ by +0.073 against +0.0028 (A). **D.** The joint phase diagram of Theorem 5.1: measured γ_{tail}^+ on the 19×19 grid, with the boundaries $\alpha_\lambda = \frac{1}{2}$ and $\alpha_\theta = \frac{1}{2}$ (dashed). The lower-right quadrant is the rotation-driven singular phase of cryptic iso-spectral allostery; the upper-right quadrant is the only equivalent phase.

6. SELF-CRITICAL ASSESSMENT

We separate what is established from what is not.

Established unconditionally. The closed-form coefficients (2.2) and the exact block formulas (2.5) and (5.2) (Propositions 2.2, 2.4, Lemma 5.2); the series representation (3.2), the three-regime asymptotics with their constants, and the spectral Feldman–Hájek dichotomy at $\alpha_c = \frac{1}{2}$ for each mechanism (Theorem 3.1); the exact cancellation of the Euler–Maclaurin constants by the doubling increment and the limit (4.4) (Theorem 4.1); the refined slope formula (4.5) (Theorem 4.2); the parity statements and the shift formula (4.8) (Theorem 4.3(1)–(3)); the joint phase criterion (5.1) (Theorem 5.1).

Conditional or incomplete. The cross-over formula of Corollary 5.3 treats the slowly varying rotational coefficient κ_k through the remainder; its full analytic form reproduces the grid to 4×10^{-4} , but the leading closed form is only accurate to 0.013 where the Euclidean amplitude shift is large. The model spectra are idealised: r is the same for every pair, the mode pairing is fixed in advance, and no mode matching problem arises. The identification of α_λ and α_θ with measurable properties of a real protein is not attempted. As in Papers I–II, $N \rightarrow \infty$ is an idealisation and every finite- N affinity is strictly positive; the theorems concern scaling with N .

Relation to the literature. The Feldman–Hájek theorem [7, 8] and Kakutani’s theorem [9] are the two dichotomies used; the affine-invariant geometry of $\mathcal{P}_N$ and its geodesic (log-Euclidean in the commuting case) coordinates are classical [3, 14]; the Euler–Maclaurin expansion of $H_N(s)$ is standard [2, 15]. Principal-component (essential-dynamics) analysis of protein covariances [1] supplies the modes on which the two mechanisms act, and the ensemble view of allostery with its cryptic sites [4, 12, 13] supplies the premise. We are not aware of a treatment of covariance-spectrum perturbations as a Feldman–Hájek critical phenomenon, nor of the doubling estimator in this context; a literature check remains advisable.

TABLE 6. Assessment of Paper III.

Item	Status	Where
Exact per-mode / per-pair contributions and their series	established	Props. 2.2, 2.4
Three-regime asymptotics, constants, $\alpha_c = \frac{1}{2}$ for A and B	established	Thm. 3.1
Agreement with data: 3×10^{-12} at $N \geq 100$	verified numerically	Tables 1, 2
Doubling tail cancels $\zeta(2\alpha)$; $\gamma_{\text{tail}} \rightarrow 1 - 2\alpha$ on $(0, 1)$	established	Thm. 4.1
Refined naive-slope formula reproduces γ_{naive} to 10^{-4}	established / verified	Thm. 4.2, Table 3
Geodesic parity; Euclidean pull-back; shift formula (4.8)	established	Thm. 4.3
$\gamma_{\text{tail}}(0.45)$: +0.1028 (A), +0.1001 (A-geo), +0.1001 (B)	verified numerically	Table 4
Joint phase criterion $\min(\alpha_\lambda, \alpha_\theta) \leq \frac{1}{2}$; 361/361 points	established / verified	Thm. 5.1, Table 5
Two-power-law cross-over on the joint grid; all four characteristic entries; 4×10^{-4} on 361 points	established / verified	Cor. 5.3
Empirical covariances: noise, mode matching, Marchenko–Pastur	open	Problem (1)

Open problems and next steps.

- (1) *Empirical covariance matrices.* A covariance estimated from T frames of a molecular-dynamics trajectory of N coordinates has, for T/N of order one, a bulk of noise eigenvalues distributed by the Marchenko–Pastur law [11, 10], and the corresponding eigenvectors are random. Both fingerprints are then contaminated exactly in the high-index tail that decides the dichotomy: the amplitude ratios η_i of bulk modes are pure noise, and the rotation angles θ_k of bulk modes are close to $\pi/2$ irrespective of the ligand. The next step is to filter the bulk (keep the eigenvalues above the Marchenko–Pastur edge $\sigma^2(1 + \sqrt{N/T})^2$, or apply shrinkage) before evaluating D_{tail} , and to determine how the decay exponents $\alpha_\lambda, \alpha_\theta$ of the retained modes can be estimated with a controlled bias.
- (2) *Mode matching.* For real spectra the pairing of modes between C_A and C_B is not given; the rotation fingerprint should be defined through an optimal assignment (or through the principal angles between the subspaces spanned by the first k modes), and Theorem 5.1 should be restated for that definition.
- (3) *The rotational coefficient.* Corollary 5.3 absorbs the k -dependence of $\kappa_k = \kappa(r; x_{1,k}, x_{2,k})$ into a remainder; an explicit expansion of $\bar{\kappa}$ in powers of $k^{-\alpha_\lambda}$ would turn the leading closed form (5.5) into a two-term formula of the same accuracy as the full analytic fit.
- (4) *Beyond fixed anisotropy.* Allow r_k to vary with k and θ_k to couple non-adjacent modes; the block formula (5.2) generalises to any 2×2 invariant subspace, but the Givens structure does not.

Reproducibility. All results are generated by two scripts in `code/`: `exp03_spectral_kakutani_criticality.py` (data, figure, acceptance tests) and `paper03_compute_tables.py` (series coefficients, Euler–Maclaurin predictions, limits, the geodesic variant, the shift formula, and the table bodies; its text summary is `results/paper03_tables_summary.txt`); the first runs in about a minute, the second in seconds, with NumPy, Matplotlib and mpmath. The paper and its L^AT_EX source (`paper/`), the raw data (`data/`), the figure (`figures/`) and the derived tables (`results/`) are archived under DOI [10.5281/zenodo.23015709](https://doi.org/10.5281/zenodo.23015709) and maintained at <https://github.com/Ruqing1963/kakutani-statistical-pharmacology-III>.

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